

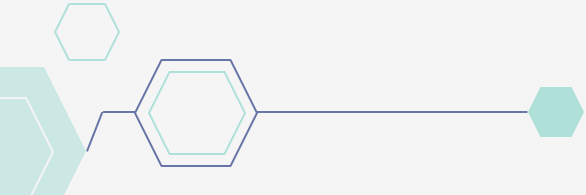
The Effect of Drug A on Fasting Sugar Levels: A Two- Way ANOVA Analysis

Examining Treatment Effectiveness Across Multiple Clinical Sites

Jiya Patel

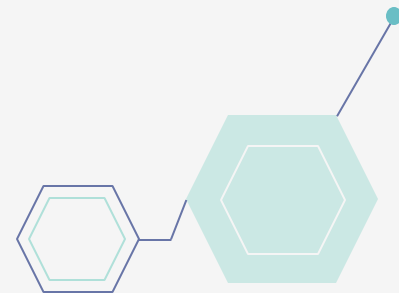
BINF 5210

December 2nd, 2025



Overview

1. Objective
2. Research Question as Hypothesis
3. Null and Alternative Hypothesis
4. T-Test/Anova Choice
5. Methodology
6. Descriptive Statistics
7. ANOVA Findings
8. Challenges
9. Conclusion with Full SAS Coding



Study Objective

Study Purpose: To evaluate the effectiveness of Drug A at different dosage levels (Low-Dose and High-Dose) compared to placebo in reducing fasting sugar levels, and to determine whether treatment effectiveness varies across clinical sites.

Study Background:

- Randomized controlled trial with 3,400 patients enrolled across 3 clinical sites
- Patients randomly assigned to three treatment groups:
 - Placebo (Control)
 - Low-Dose Drug A
 - High-Dose Drug A
- 10-week treatment period
- Primary outcome: Test Score (Fasting Sugar Level)

Research Question

Research Question: Is there a significant difference in mean fasting sugar levels among the three treatment groups (Low-Drug A, High-Drug A, Placebo) across all three sites, and does the site location influence treatment effectiveness?

*** There will be three sets of hypotheses with this question.***

Hypothesis: Main Effect of Group

Null Hypothesis: There is no significant difference in mean fasting sugar levels among the three treatment groups (Placebo, Low-Dose Drug A, and High-Dose Drug A).

$$\mu_{\text{Placebo}} = \mu_{\text{Low}} = \mu_{\text{High}}$$

Alternative Hypothesis: There is a significant difference in mean fasting sugar levels among at least two of the three treatment groups.

$$\text{Significance Level} = \alpha = 0.05$$

Hypothesis: Main Effect of Site

Null Hypothesis: There is no significant difference in mean fasting sugar levels among the three sites (Site 1, Site 2, Site 3).

$$\mu_{\text{Site1}} = \mu_{\text{Site2}} = \mu_{\text{Site3}}$$

Alternative Hypothesis: There is a significant difference in mean fasting sugar levels among at least two of the three sites.

$$\text{Significance Level} = \alpha = 0.05$$

Hypothesis: Interaction Effect

Null Hypothesis: There is no interaction between treatment group and site location on mean fasting sugar levels.

The effect of treatment is the same across all sites.

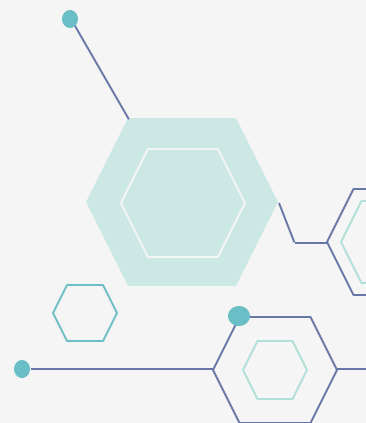
Alternative Hypothesis: There is a significant interaction between treatment group and site location on mean fasting sugar levels.

The effect of treatment varies depending on which site patients were treated at.

Significance Level = $\alpha = 0.05$

A decorative graphic in the top-left corner consisting of several overlapping hexagons. Some hexagons are solid light green, while others are outlined in a thin blue line. Small blue dots are scattered around the hexagons.

Why Two-Way ANOVA Test?

A decorative graphic in the bottom-right corner featuring a large solid light green hexagon with a blue outline. It is connected by thin blue lines to other hexagons, including one with a blue dot and another with a blue outline. Small blue dots are also present near the connections.

Why These Test will not work?

<u>One sample T-Test</u>	There is no fixed hypothesis given in the assignment prompt.
<u>Two sample T-Test</u>	There is no categorical variable of exactly two groups. The dataset has three categories in Group.
<u>Paired T-Test</u>	The dataset given provides post-treatment scores after 10 weeks. There is no baseline measurements.
<u>One Way ANOVA</u>	Can't detect interactions and doesn't fit the assignment requirements.

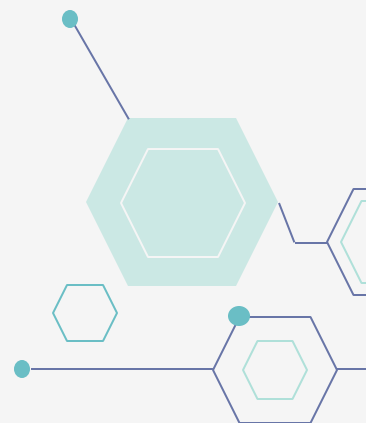
Why Two-Way ANOVA works:

- ✓ Two categorical independent variables (Group with 3 levels, Site with 3 levels)
- ✓ One continuous dependent variable (Test Score)
- ✓ Addresses potential confounding
- ✓ Can test for interaction
- ✓ Explains “difference among groups across all three sites”

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Methodology

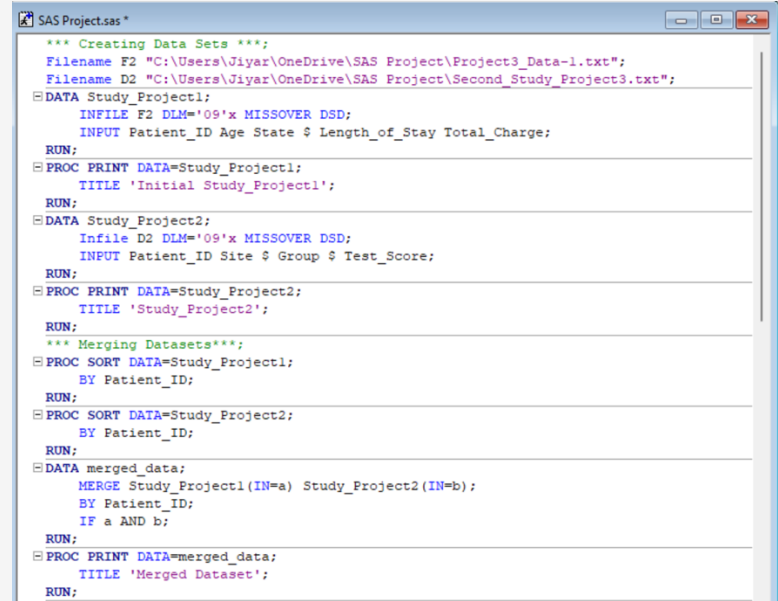
Data Integration, Random Sampling,
& Statistical Analysis

A decorative graphic in the bottom-right corner featuring a large solid light green hexagon with a dark blue outline. It is connected by thin dark blue lines to other hexagons, including one outlined in light green and another outlined in dark blue. Small dark blue dots are also present.



Data Integration

- Merged two datasets on Patient_ID
- Dataset 1: Demographics (Age, State, Length_of_Stay, Total_Charge)
- Dataset 2: Treatment info (Site, Group, Test_Score)
- Result: N = 3,400 complete records



```
SAS Project.sas

*** Creating Data Sets ***;
Filename F2 "C:\Users\Jiyar\OneDrive\SAS Project\Project3_Data-1.txt";
Filename D2 "C:\Users\Jiyar\OneDrive\SAS Project\Second_Study_Project3.txt";

DATA Study_Project1;
  INFILE F2 DLM='09'x MISSOVER DSD;
  INPUT Patient_ID Age State $ Length_of_Stay Total_Charge;
RUN;

PROC PRINT DATA=Study_Project1;
  TITLE 'Initial Study_Project1';
RUN;

DATA Study_Project2;
  Infile D2 DLM='09'x MISSOVER DSD;
  INPUT Patient_ID Site $ Group $ Test_Score;
RUN;

PROC PRINT DATA=Study_Project2;
  TITLE 'Study_Project2';
RUN;

*** Merging Datasets***;
PROC SORT DATA=Study_Project1;
  BY Patient_ID;
RUN;

PROC SORT DATA=Study_Project2;
  BY Patient_ID;
RUN;

DATA merged_data;
  MERGE Study_Project1(IN=a) Study_Project2(IN=b);
  BY Patient_ID;
  IF a AND b;
RUN;

PROC PRINT DATA=merged_data;
  TITLE 'Merged Dataset';
RUN;
```

SAS Output

Picture shows "OBS=10" simply for powerpoint use, the full N=1000 was used for the analysis.

Initial Study_Project1

Obs	Patient_ID	Age	State	Length_of_Stay	Total_Charge
1	1	51	AZ	1	3170
2	2	50	AZ	2	4319
3	3	74	AZ	3	5850
4	4	59	AZ	1	6298
5	5	57	AZ	5	5386
6	6	71	AZ	4	4737
7	7	73	AZ	3	7525
8	8	49	AZ	1	2817
9	9	30	AZ	2	3430
10	11	20	AZ	2	3316

Study_Project2

Obs	Patient_ID	Site	Group	Test_Score
1	1	Site 3	Low	46.6496
2	2	Site 2	High	35.3573
3	3	Site 1	Placebo	87.6297
4	4	Site 2	Low	44.5998
5	5	Site 1	Low	59.0460
6	6	Site 1	High	77.4058
7	7	Site 1	Placebo	65.6273
8	8	Site 1	High	49.4666
9	9	Site 1	Low	42.7179
10	10	Site 1	High	83.7879

Merged Dataset

Obs	Patient_ID	Age	State	Length_of_Stay	Total_Charge	Site	Group	Test_Score
1	1	51	AZ	1	3170	Site 3	Low	46.6496
2	2	50	AZ	2	4319	Site 2	High	35.3573
3	3	74	AZ	3	5850	Site 1	Placebo	87.6297
4	4	59	AZ	1	6298	Site 2	Low	44.5998
5	5	57	AZ	5	5386	Site 1	Low	59.0460
6	6	71	AZ	4	4737	Site 1	High	77.4058
7	7	73	AZ	3	7525	Site 1	Placebo	65.6273
8	8	49	AZ	1	2817	Site 1	High	49.4666
9	9	30	AZ	2	3430	Site 1	Low	42.7179
10	11	20	AZ	2	3316	Site 1	Placebo	35.3631



Random Sampling

- **Method:** Simple Random Sampling (SRS)
- **Sample Size:** N = 1,000 from 3,400 patients
- **SAS Code:** PROC SURVEYSELECT METHOD=SRS N=1000
- **No seed used** (true randomization)

```
*** Simple Random Sampling ***;
Proc Surveyselect data = merged_data
Method=SRS
N=1000
OUT=Sample_Data;
Run;

Proc Print Data = Sample_Data;
Title 'SRS (N=1000)';
Run;

*** Descriptive Statistics ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
Var Age Length_of_Stay Total_Charge Test_Score;
Title 'Descriptive Statistics - Full Sample (N=1000)';
Run;

*** Descriptive Statistics by Group ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
Class Group;
Var Test_Score;
Title 'Test Score by Treatment Group (Sample Data)';
Run;

*** Descriptive Statistics by Site ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
Class Site;
Var Test_Score;
Title 'Test Score by Site (Sample Data)';
Run;

*** Descriptive Statistics by Group and Site ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
Class Group Site;
Var Test_Score;
Title 'Test Score by Group and Site (Sample Data)';
Run;
```



SAS Output

Of the first 15 SRS samples

Merged Dataset	
The SURVEYSELECT Procedure	
Selection Method	Simple Random Sampling
Input Data Set	MERGED_DATA
Random Number Seed	572429000
Sample Size	1000
Selection Probability	0.312891
Sampling Weight	3.196
Output Data Set	SAMPLE_DATA

SRS (N=1000)

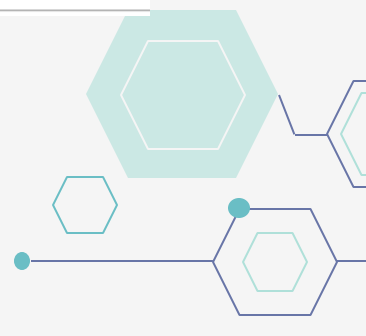
Obs	Patient_ID	Age	State	Length_of_Stay	Total_Charge	Site	Group	Test_Score
1	9	30	AZ	2	3430	Site 1	Low	42.718
2	11	20	AZ	2	3316	Site 1	Placebo	35.363
3	14	29	AZ	1	2493	Site 2	Placebo	33.087
4	41	82	AZ	4	7518	Site 1	High	133.163
5	42	15	AZ	1	2051	Site 1	Placebo	43.959
6	45	31	AZ	3	5371	Site 1	Placebo	52.300
7	47	33	AZ	1	1612	Site 2	n/a	31.562
8	53	80	AZ	5	9991	Site 3	High	89.838
9	58	29	AZ	2	4694	Site 1	Placebo	92.664
10	59	82	AZ	3	4576	Site 1	Placebo	80.420



SAS Code

For Descriptive Statistics of all Hypotheses

```
*** Descriptive Statistics ***;  
Proc Means Data = Sample_Data N Mean Std Min Max;  
  Var Age Length_of_Stay Total_Charge Test_Score;  
  Title 'Descriptive Statistics - Full Sample (N=1000)';  
Run;  
*** Descriptive Statistics by Group ***;  
Proc Means Data = Sample_Data N Mean Std Min Max;  
  Class Group;  
  Var Test_Score;  
  Title 'Test Score by Treatment Group (Sample Data)';  
Run;  
*** Descriptive Statistics by Site ***;  
Proc Means Data = Sample_Data N Mean Std Min Max;  
  Class Site;  
  Var Test_Score;  
  Title 'Test Score by Site (Sample Data)';  
Run;  
*** Descriptive Statistics by Group and Site ***;  
Proc Means Data = Sample_Data N Mean Std Min Max;  
  Class Group Site;  
  Var Test_Score;  
  Title 'Test Score by Group and Site (Sample Data)';  
Run;
```



Descriptive Statistics – Key Findings

Normality Assessment:

- Mean > Median = Skewed
- Kurtosis is higher than 3
- Extreme right skewed

Key Takeaways:

- VERY high variability (Std. Dev)
- Data is not normally distributed.
- R^2 = Treatment group and site are **weak predictors** of fasting sugar levels

Statistics	PROC UNIVARIATE
Sample Size	1000
Mean	99.75
Median	69.91
Std. Dev.	89.49
Kurtosis	66.37
Skewness	5.52
R^2	0.031180

Full SAS Output

Descriptive Statistics - Full Sample (N=1000)

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
Age	989	54.9292214	19.1255453	8.0000000	100.0000000
Length_of_Stay	1000	4.8990000	5.6155439	0	67.0000000
Total_Charge	929	24817.45	35963.80	995.0000000	419956.00
Test_Score	1000	99.7532627	89.4867953	22.7021500	1521.49

Test Score by Site (Sample Data)

The MEANS Procedure

Analysis Variable : Test_Score						
Site	N Obs	N	Mean	Std Dev	Minimum	Maximum
N/a	1	1	31.9845300	.	31.9845300	31.9845300
Site 1	470	470	102.1823746	80.5547699	22.7021500	539.4668600
Site 2	338	338	96.9602764	76.6387621	28.6007100	528.3965500
Site 3	191	191	99.0732389	124.9640803	24.6664900	1521.49

Test Score by Group and Site (Sample Data)

The MEANS Procedure

Analysis Variable : Test_Score							
Group	Site	N Obs	N	Mean	Std Dev	Minimum	Maximum
High	Site 1	104	104	103.9360303	77.1298843	29.0494400	402.3336200
	Site 2	86	86	104.0442924	80.1538622	35.6090600	414.6664400
	Site 3	43	43	75.0996612	36.4598416	25.1185000	175.2972000
Low	Site 1	121	121	84.6510781	65.7318355	25.1107400	435.9954500
	Site 2	81	81	87.2571175	67.6358501	34.6065900	408.7367600
	Site 3	57	57	91.1488093	76.9557017	25.2057200	362.2584200
Placebo	Site 1	186	186	112.7154730	86.9804766	22.7021500	504.0502000
	Site 2	122	122	98.5798735	78.6052669	28.6007100	508.3172900
	Site 3	69	69	95.1633796	73.6169843	24.6664900	368.5328100
n/a	N/a	1	1	31.9845300	.	31.9845300	31.9845300
	Site 1	59	59	101.8391605	88.5083401	24.5216700	539.4668600
	Site 2	49	49	96.5345551	80.0262971	31.4939100	528.3965500
	Site 3	22	22	178.7249036	311.8396259	36.0882300	1521.49

Test Score by Treatment Group (Sample Data)

The MEANS Procedure

Analysis Variable : Test_Score						
Group	N Obs	N	Mean	Std Dev	Minimum	Maximum
High	233	233	98.6542564	73.2241473	25.1185000	414.6664400
Low	259	259	86.8960969	68.7017746	25.1107400	435.9954500
Placebo	377	377	104.9286359	82.1675867	22.7021500	508.3172900
n/a	131	131	112.2338632	150.0265680	24.5216700	1521.49

ANOVA Test Results

PROC GLM PROCEDURE			
Source	Type III SS	P-Value	Decision?
Group	115105.66	0.0022	Reject Null
Site	27017.45	0.3291	Fail to Reject
Group*Site	163411.25	0.0021	Reject Null

```
*** Normality Test***;
Proc Univariate Data = Sample_Data Normal Plot;
  Var Test_Score;
  Histogram Test_Score / Normal;
  Title 'Normality Test for Test_Score (Univariate Analysis)';
Run;

*** Two Way Anova Test ***;
Proc GLM Data = Sample_data;
  Class Group Site;
  Model Test_score = Group Site Group*Site;
  Means Group Site;
  Title "With values";
Run;
Quit;
```

Treatment Group effect was significant ($p = 0.0022$):

- Mean fasting sugar levels differed between at least two treatment groups.

Site effect was not significant ($p = 0.3291$)

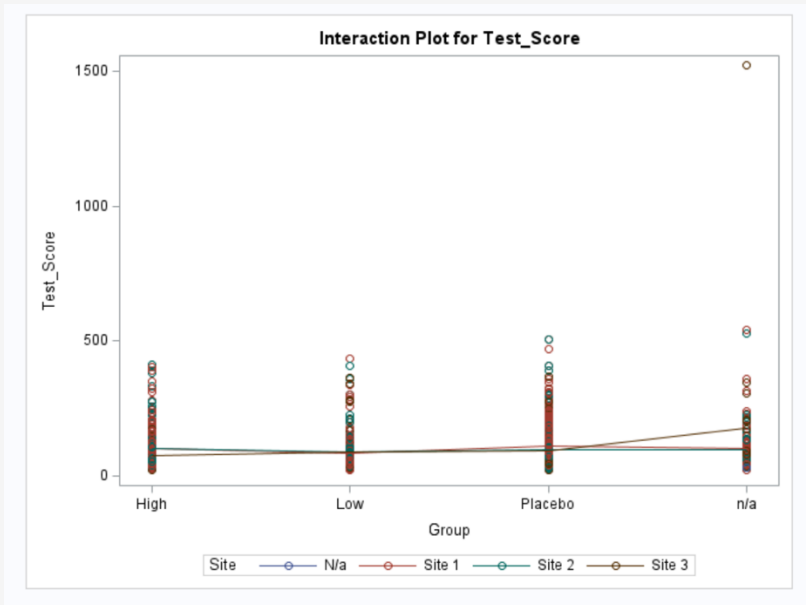
- Fasting sugar levels were similar across the three locations.

Group x Site interaction was significant ($p = 0.0021$)

- Treatment effectiveness varied by *site*.

Interaction Plot

Since they're not parallel, it means treatment effects differ by site





Challenges

- Extreme Non-Normality: Severe right skew
- Outlier is present (~1500)
- High Variability
- Random Sampling Variability → No seed used → Results may vary across samples
- 10-week timeframe may not capture long-term treatment effects
- Statistical significance $\neq$ Practical significance





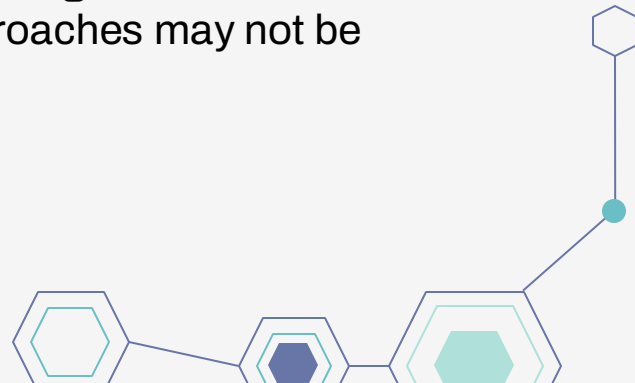
Summary

Clinical Significance:

While effects are statistically significant, they explain only 3.1% of variance ($R^2 = 0.031$). Individual patient factors such as genetics, lifestyle, baseline health are far more important than treatment assignment in determining fasting sugar levels.

Key Takeaway:

Drug A shows promise but requires site-specific implementation strategies. The significant interaction highlights the importance of considering local factors in treatment delivery and suggests that one-size-fits-all approaches may not be optimal.



Full SAS Code

```
SAS Project.sas

*** Creating Data Sets ***;
Filename F2 "C:\Users\Jiyar\OneDrive\SAS Project\Project3_Data-1.txt";
Filename D2 "C:\Users\Jiyar\OneDrive\SAS Project\Second_Study_Project3.txt";

DATA Study_Project1;
  INFILE F2 DLM='09'x MISSOVER DSD;
  INPUT Patient_ID Age State $ Length_of_Stay Total_Charge;
RUN;

PROC PRINT DATA=Study_Project1;
  TITLE 'Initial Study_Project1';
RUN;

DATA Study_Project2;
  Infile D2 DLM='09'x MISSOVER DSD;
  INPUT Patient_ID Site $ Group $ Test_Score;
RUN;

PROC PRINT DATA=Study_Project2;
  TITLE 'Study_Project2';
RUN;

*** Merging Datasets***;
PROC SORT DATA=Study_Project1;
  BY Patient_ID;
RUN;

PROC SORT DATA=Study_Project2;
  BY Patient_ID;
RUN;

DATA merged_data;
  MERGE Study_Project1(IN=a) Study_Project2(IN=b);
  BY Patient_ID;
  IF a AND b;
RUN;

PROC PRINT DATA=merged_data;
  TITLE 'Merged Dataset';
RUN;
```

```
*** Simple Random Sampling ***;
Proc Surveyselect data = merged_data
  Method=SRS
  N=1000
  OUT=Sample_Data;
Run;

Proc Print Data = Sample_Data;
  Title 'SRS (N=1000)';
Run;

*** Descriptive Statistics ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
  Var Age Length_of_Stay Total_Charge Test_Score;
  Title 'Descriptive Statistics - Full Sample (N=1000)';
Run;

*** Descriptive Statistics by Group ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
  Class Group;
  Var Test_Score;
  Title 'Test Score by Treatment Group (Sample Data)';
Run;

*** Descriptive Statistics by Site ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
  Class Site;
  Var Test_Score;
  Title 'Test Score by Site (Sample Data)';
Run;

*** Descriptive Statistics by Group and Site ***;
Proc Means Data = Sample_Data N Mean Std Min Max;
  Class Group Site;
  Var Test_Score;
  Title 'Test Score by Group and Site (Sample Data)';
Run;
```

```
*** Normality Test***;
Proc Univariate Data = Sample_Data Normal Plot;
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  Histogram Test_Score / Normal;
  Title 'Normality Test for Test_Score (Univariate Analysis)';
Run;

*** Two Way Anova Test ***;
Proc GLM Data = Sample_data;
  Class Group Site;
  Model Test_score = Group Site Group*Site;
  Means Group Site;
  Title "With values";
Run;
Quit;
```

Thank You!



Do you have any questions?

jp2705@rutgers.edu

I enjoyed testing my analytical skills while
obtaining hands on data cleaning,
descriptive statistics, and data
visualization.

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